

Operator Discovery Report: scaffold_selector_structure_aware

- Operator type: scaffold_selector.
- Host scaffold: nearest_neighbor_2opt.
- Core idea: Deterministically select a best-fit baseline scaffold from instance descriptors (two-cluster bottleneck, clustered/grid/elongated, and nearest-neighbor trap), otherwise fall back to nearest_neighbor_2opt.
- Novelty classification: descriptor_based_scaffold_selector.
- Rediscovery signature: scaffold_selector:random_restart_2opt:clustered_local_search:sparse_three_opt.
- Surviving candidate: False.
- Problem class where it helps: heldout_tsplib, clustered_tsp, two_cluster_bottleneck_tsp.

Pseudocode

- Read instance descriptors to classify the layout family.
- Dispatch the instance to the preferred scaffold for that structure class.
- Fall back to a safe default scaffold when the structure is ambiguous.

Complexity

- Non-empty lines: 13.
- Complexity score: 0.46.
- Fingerprint: b694c0a51632de6b.

Why It Should Help

- The operator selects among simple solvers from instance descriptors, so any gain should come from a reusable structure-to-scaffold mapping.

Failure Cases

- Underperforms on grid_like_tsp with gap delta 0.170114.
- Underperforms on uniform_euclidean with gap delta 0.013605.
- Does not transplant cleanly into cheapest_insertion_2opt.
- Runtime inflation is high at 1.442123.

Transplant Results

- nearest_neighbor_2opt gap delta -0.057811, runtime delta 335.973914, distance-eval delta 109924.428571.
- cheapest_insertion_2opt gap delta 0.049511, runtime delta -3708.730271, distance-eval delta -6651133.571429.
- random_restart_2opt gap delta -0.037485, runtime delta 103.263586, distance-eval delta 94655.428571.
- sparse_three_opt gap delta -0.056525, runtime delta 327.453757, distance-eval delta 106503.142857.
- clustered_local_search gap delta -0.0842, runtime delta 178.8486, distance-eval delta 97304.142857.

Instance-Family Results

- heldout_tsplib operator gap 0.176732 vs baseline 0.235058.
- clustered_tsp operator gap 0.038868 vs baseline 0.110853.
- elongated_corridor_tsp operator gap 0.006647 vs baseline 0.003204.
- grid_like_tsp operator gap 0.27681 vs baseline 0.106696.
- nearest_neighbor_trap_tsp operator gap 0.0 vs baseline 0.0.
- two_cluster_bottleneck_tsp operator gap 0.001685 vs baseline 0.109057.
- uniform_euclidean operator gap 0.013605 vs baseline 0.0.

Pareto Results

- Gap delta: -0.019489.
- Runtime inflation: 1.442123.
- Distance-evaluation inflation: 11.648842.
- Same-gap-faster flag: False.
- Better-gap-same-runtime flag: False.

Ablation Results

- disabled mean gap 0.121317 and delta vs full operator 0.019114.
- flattened_selector mean gap 0.121317 and delta vs full operator 0.019114.
- shuffled_selector mean gap 0.073227 and delta vs full operator -0.028976.